

DECKA FADHILA TIRTA

Bachelor of Science in Physics | Data | Research | Technology | Operations

deckafadhila@gmail.com | Indonesia

LinkedIn: deckafadhila | GitHub: dkfadhila | X: @tirtavex | dktirta.tech

PROFESSIONAL SUMMARY

Bachelor of Science in Physics graduate from Universitas Negeri Yogyakarta with experience in data processing, scientific computing, research, administrative support, and organizational coordination. Experienced in handling large datasets, preparing reports and documentation, supporting academic activities, and developing computational projects using Python. Skilled in Microsoft Office, Google Workspace, data visualization, and digital productivity tools. Strong interest in research and technology, with a highly curious and adaptable approach to learning new tools and solving problems.

CORE COMPETENCIES

Data Management & Processing | Data Analysis & Visualization | Scientific Computing | Research & Information Analysis
Reporting & Documentation | Administrative Support | Workflow & Process Management | Data Quality & Organization
Microsoft Office & Google Workspace | AI-Leveraged Operations & Automation | Project & Task Planning | Content Operations
Graphic Design & Visual Communication | Social Media Management | Copywriting & Content Writing

AI & AUTOMATION

- Use AI tools and automation to improve research, content creation, information analysis, and repetitive workflows.
 - Develop and experiment with research workflows, knowledge bases, content operations, agent workflows, and workflow automation.
 - Experienced with n8n, Cursor, GitHub, Claude Code, Codex, Vercel, and Hermes.
- AI Tools:** ChatGPT (reasoning, research, analysis), Claude (research, concept development), Gemini (multimodal research), Grok (real-time information), Hermes (agentic workflows)

EDUCATION

Bachelor of Science in Physics (S.Si.) Graduated 2026
Universitas Negeri Yogyakarta

PROFESSIONAL EXPERIENCE

Digital Systems Laboratory Teaching Assistant Aug 2024 - Dec 2024
Universitas Negeri Yogyakarta

- Processed and managed assessment data for 20+ students during Digital Systems laboratory sessions.
- Supported student assessment and documentation processes.
- Assisted in monitoring students during laboratory activities.
- Provided guidance to students in completing Digital Systems laboratory assignments.
- Supported coordination between students and teaching staff.

Intern - Emergency & Logistics Division Dec 2024 - Feb 2025
Badan Penanggulangan Bencana Daerah (BPBD) Kabupaten Kudus

- Sorted and organized travel documents and financial and expense records for the Emergency and Logistics Division.
- Supported administrative document management and information organization.
- Created visual materials and informational designs for BPBD communication needs.
- Conducted rainfall and disaster risk and vulnerability mapping for Kabupaten Kudus.
- Participated in field surveys and field assessments related to landslides and flood events.
- Assisted in collecting and organizing field information related to disaster conditions.

Physics Computing Teaching Assistant Feb 2025 - Jun 2025
Universitas Negeri Yogyakarta

- Processed and managed student assessment data for 60+ students across laboratory and theoretical assessments.
- Organized student assessment data to support accurate grading and academic reporting.
- Assisted lecturers and students during Physics Computing laboratory sessions.
- Guided students in completing computational physics assignments and laboratory activities.
- Provided technical guidance and troubleshooting support during laboratory sessions.

SELECTED PROJECTS

Project Veyra - Multiplayer Browser RPG (Realm of the Soulbond) 2026

Full-Stack Web Game Project | github.com/dkfadhila/ProjectVeyra

- Built a full-stack 2D browser RPG: vanilla JavaScript + Canvas2D frontend, Express.js + Socket.IO backend with real-time multiplayer position sync.
- Implemented JWT + bcrypt authentication, auto-save persistence (server and localStorage), and on-chain verified player state (OG Storage).
- Developed a player-driven marketplace with token-based economy, turn-based combat with skills and item drops, chained quest system, and 7 playable classes.
- Integrated an AI-powered NPC (Lyra Oracle, MIMO) responding to player dialogue and gifts.
- Developed end-to-end with an AI-leveraged workflow using Claude Code.

Simulating the Orbits of 83 Saturnian Moons Using Python & Pygame

Computational Physics Project

- Processed and organized 30,000+ data points related to 83 known Saturnian moons.
- Worked with orbital parameters including velocity, period, position, trajectory, and mass.
- Developed a computational simulation to model the movement and orbital behavior of Saturnian moons.
- Applied the 4th-order Runge-Kutta method to predict satellite positions and velocities.
- Modeled gravitational interactions based on Newton's Law of Gravitation and Kepler's Laws.

Tools: Python, Pygame, Scientific Computing, Data Processing, Numerical Methods

Analysis of the March 25, 2025 Flood Event in Kabupaten Kudus

Atmospheric Dynamics Analysis

- Conducted atmospheric dynamics analysis using Python to investigate factors associated with the flood event.
- Processed and analyzed 10,000+ data points covering the period from 2016 to 2025.
- Analyzed rainfall, surface temperature, atmospheric pressure, cloud movement, and climate variability.

Tools: Python, Data Analysis, Scientific Computing, Data Visualization, Research

Ultrasonic Sensor-Based Automatic Garage System

Embedded Systems Project

- Developed an automatic garage prototype using Arduino UNO and an ultrasonic sensor.
- Analyzed mathematical equations to calculate distance and determine system operating conditions.
- Integrated sensor input, mathematical calculations, and Arduino-based control logic.

Tools: Arduino UNO, Ultrasonic Sensor, Embedded Systems, Control Logic

ORGANIZATIONAL EXPERIENCE

Media & Information Staff - UKM CATUR UNY

2023 - 2025

- Supported internal communication and coordination within the Media & Information division.
- Managed and published organizational social media content.
- Created visual materials for organizational communication.
- Supported content planning and publication activities.

Organizational Advisory Board - UKM CATUR UNY

2025

- Monitored internal coordination and organizational relationships among management members.
- Provided organizational monitoring and oversight to support internal coordination.

Vice Head - Media & Information Division - BEM FMIPA UNY

2025

- Coordinated communication between organizational leadership, Media & Information staff, and other divisions.
- Monitored workflow and coordination within the Media & Information division.
- Managed communication and coordination related to organizational media activities.
- Created visual content and promotional materials for organizational needs.
- Supported social media publication and content distribution.
- Handled design requests from different organizational divisions.

WEB3 EXPERIENCE

Independent Web3 Researcher & Ecosystem Participant

2024 - Present

- Researched and evaluated Web3 projects based on project activity, ecosystem development, incentive mechanisms, product concepts, and community participation.
- Participated in and contributed to 300+ Web3 project campaigns and ecosystem activities through social tasks, testnets, and community activities.
- Developed practical understanding of blockchain technology, DeFi, decentralized exchanges, liquidity pools, DLMM, token ecosystems, and Web3 protocols.
- Monitored project updates, product developments, ecosystem narratives, campaigns, and community activities.

Web3 Content Writer & Researcher - X: @tirtavex

2024 - Present

- Researched and wrote content related to Web3 projects, blockchain protocols, DeFi concepts, and ecosystem developments.
- Translated technical or project-related information into accessible written content.
- Created copywriting for Web3-related social media content.

TECHNICAL & DATA SKILLS

Data & Research: Data Processing, Data Management, Data Analysis, Data Visualization, Scientific Computing, Numerical Methods, Research & Information Analysis, Large Dataset Handling, Python

Office & Productivity: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Google Sheets, Google Docs, Google Slides, Google Workspace, Notion, Documentation & Reporting

AI & Automation: AI Agents, AI-Leveraged Operations, Workflow Automation, Research Workflows, Knowledge Bases, Agent Workflows, n8n

Embedded Systems: Arduino UNO, Sensor-Based Automation, Ultrasonic Sensors, Basic Electronics, Control Logic

Design & Content: Canva, Figma, Adobe Photoshop, CapCut, Filmora, DaVinci Resolve, Graphic Design, Social Media Visuals, Content Writing, Copywriting, Video Editing

ADDITIONAL STRENGTHS

Analytical Thinking | Problem Solving | Research-Oriented Mindset | Intellectual Curiosity | Independent Learning | Adaptability | Attention to Detail | Data-Driven Mindset | Cross-Functional Communication | Emerging Technology Research | Workflow Optimization

LANGUAGES

Indonesian - Native

English - Basic-Intermediate; able to understand technical and general contexts

PROFESSIONAL INTERESTS

Artificial Intelligence | Data & Analytics | Scientific Computing | Research | Automation | Emerging Technologies | Technology & Product Development | Web3